

## L3.2 Adversarial Attacks Quiz ANS

### 1. What is distribution shift in supervised machine learning?

- A) The inference-data distribution differs from the training-data distribution
- B) The model always receives exactly the training data
- C) The optimizer changes from gradient descent to random search
- D) The class labels are alphabetically reordered

ANS: A) The inference-data distribution differs from the training-data distribution

### 2. In the panda adversarial-example demonstration, what does the small perturbation cause the model to predict with high confidence?

- A) A stop sign
- B) A gibbon
- C) A pedestrian
- D) A panda with lower resolution

ANS: B) A gibbon

### 3. What is the main purpose of constraining the norm of an adversarial perturbation?

- A) To guarantee a larger neural network
- B) To maximize the image resolution
- C) To limit the size of the input modification
- D) To remove all model gradients

ANS: C) To limit the size of the input modification

### 4. Which perturbation pattern is most closely associated with changing every pixel by a small amount?

- A) Changing only the class label
- B) Changing one pixel by an unlimited amount
- C) Deleting the entire image
- D) An L-infinity-bounded perturbation

ANS: D) An L-infinity-bounded perturbation

### 5. What objective is typically maximized when generating an untargeted adversarial example by local search?

- A) The model loss for the correct class
- B) The training-set size
- C) The number of output classes
- D) The image dimensions

ANS: A) The model loss for the correct class

**6. What does Projected Gradient Descent do after taking a gradient step?**

- A) It permanently enlarges the permitted perturbation region
- B) It projects the result back into the allowed perturbation set
- C) It retraines the classifier from scratch
- D) It removes every modified pixel

ANS: B) It projects the result back into the allowed perturbation set

**7. Which information does FGSM use to determine the perturbation direction?**

- A) The sign of the input image
- B) A randomly selected class label
- C) The sign of the gradient of the loss with respect to the input
- D) Only the model's training accuracy

ANS: C) The sign of the gradient of the loss with respect to the input

**8. How does PGD with small steps differ from a single-step FGSM attack?**

- A) PGD never uses gradients
- B) PGD changes only model parameters
- C) PGD cannot enforce a perturbation constraint
- D) PGD performs multiple iterative gradient steps with projection

ANS: D) PGD performs multiple iterative gradient steps with projection

**9. What is the goal of an untargeted adversarial attack?**

- A) To cause any incorrect classification
- B) To force one specific attacker-chosen label only
- C) To improve correct-class confidence
- D) To compress the input image

ANS: A) To cause any incorrect classification

**10. What is the goal of a targeted adversarial attack?**

- A) To preserve the original prediction
- B) To make the model predict a particular attacker-chosen class
- C) To increase accuracy for every class
- D) To remove the classifier's output layer

ANS: B) To make the model predict a particular attacker-chosen class

**11. What distinguishes a physically realizable adversarial attack from direct pixel manipulation?**

- A) It requires access to every model parameter
- B) It is performed only during model training
- C) It modifies or presents a physical object so that a sensor captures the attack
- D) It cannot affect traffic-sign recognition

ANS: C) It modifies or presents a physical object so that a sensor captures the attack

**12. Why must a physical adversarial example be robust to spatial transformations?**

- A) To increase the number of classes
- B) To eliminate the need for a camera
- C) To guarantee access to model weights
- D) It may be viewed from different distances, angles, and positions

ANS: D) It may be viewed from different distances, angles, and positions

**13. What knowledge is assumed in a white-box adversarial attack?**

- A) The neural-network model parameters
- B) Only the vehicle's paint color
- C) No information about the model or outputs
- D) Only the number of road lanes

ANS: A) The neural-network model parameters

**14. If an attacker has the target model's training dataset but not its parameters, what black-box strategy is described?**

- A) Disable all input sensors
- B) Train a proxy white-box model and generate attacks against it
- C) Replace the target model's hardware
- D) Use only naturally occurring distribution shifts

ANS: B) Train a proxy white-box model and generate attacks against it

**15. How can an attacker collect data for a proxy model when the target training dataset is unavailable?**

- A) By reading hidden model parameters directly
- B) By ignoring the target model
- C) By querying the target service to collect input-output pairs
- D) By changing the target's loss function

ANS: C) By querying the target service to collect input-output pairs

**16. What does the YOLO-to-Faster-RCNN example demonstrate?**

- A) Adversarial examples never transfer between models
- B) Only digital attacks affect object detectors
- C) A proxy model must be identical to the target model
- D) Physical adversarial examples may transfer from a proxy model to a black-box model

ANS: D) Physical adversarial examples may transfer from a proxy model to a black-box model

**17. What is a phantom attack against an ADAS?**

- A) A depthless displayed or projected image intended to be treated as a real 3D object
- B) A mechanical failure of the steering actuator
- C) A hidden replacement of the vehicle's processor
- D) A normal reflection that never triggers a response

ANS: A) A depthless displayed or projected image intended to be treated as a real 3D object

**18. Which delivery mechanism for a phantom attack is mentioned in the slides?**

- A) A wired keyboard inside the vehicle
- B) A drone-mounted projector or a hacked roadside digital billboard
- C) A replacement tire
- D) A paper map placed in the trunk

ANS: B) A drone-mounted projector or a hacked roadside digital billboard

**19. What training approach is used to improve robustness against adversarial inputs?**

- A) Training only on empty images
- B) Removing the loss function
- C) Including adversarially generated examples in the training optimization
- D) Freezing every model parameter before training

ANS: C) Including adversarially generated examples in the training optimization

**20. According to the slides, why is a robust model harder to attack with small gradient steps?**

- A) It has no classification output
- B) It always rejects every input
- C) It uses an infinitely large training set
- D) It has a smoother loss surface

ANS: D) It has a smoother loss surface